how a scientific paper is built

prof. dr. nicolas meseth

1 one shape for all papers
2 what goes where
3 the most common mistake
4 the abstract

readers don't read a paper from start to end, they jump to the room they need, so every room has to be where they expect it.

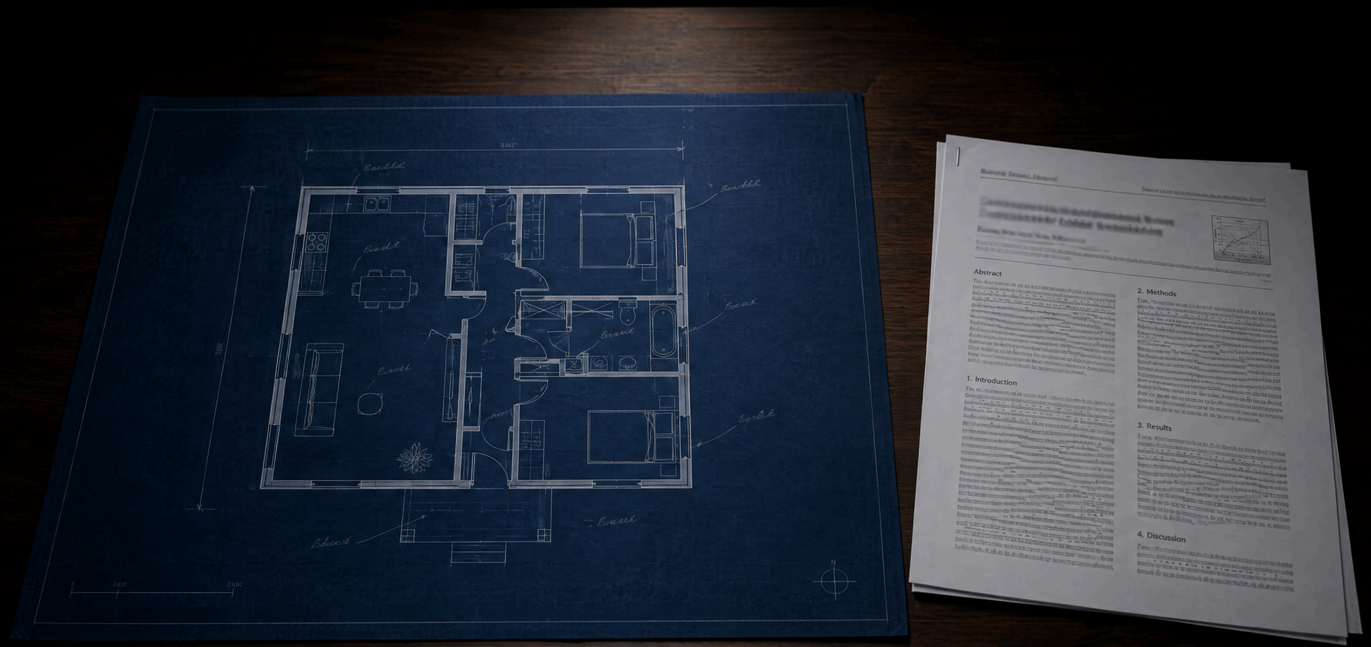


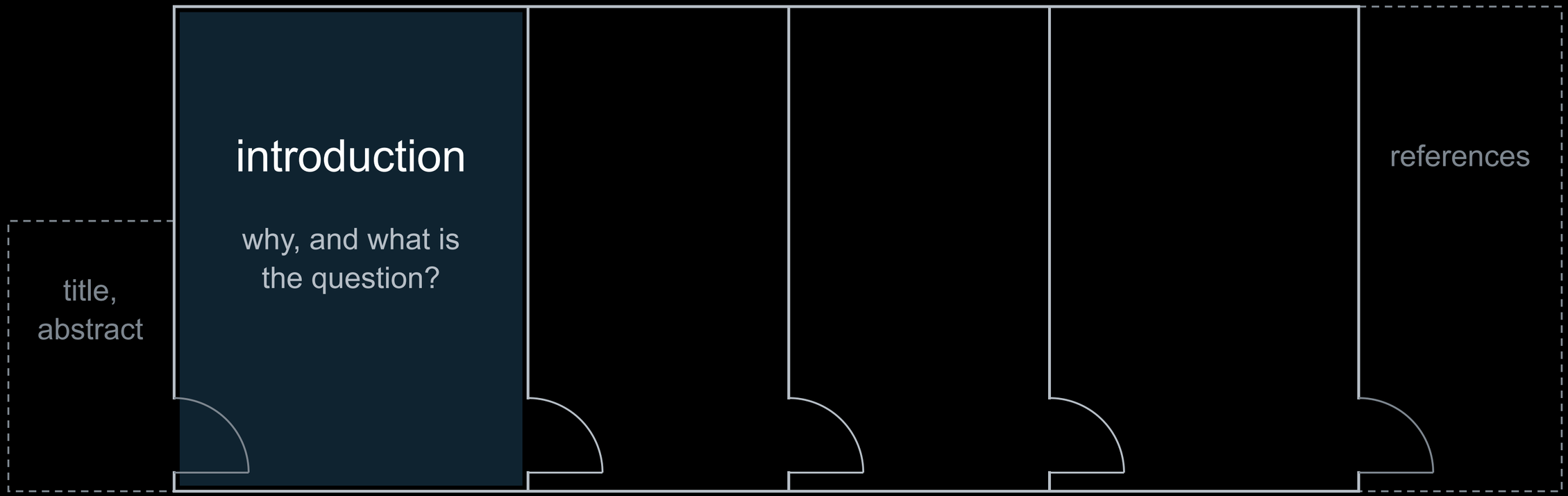
image: ai-generated (gpt-image-2.5-flare)

part 1

one shape for all papers

imrad and its proportions

imrad



one shape for all papers

imrad



one shape for all papers

imrad

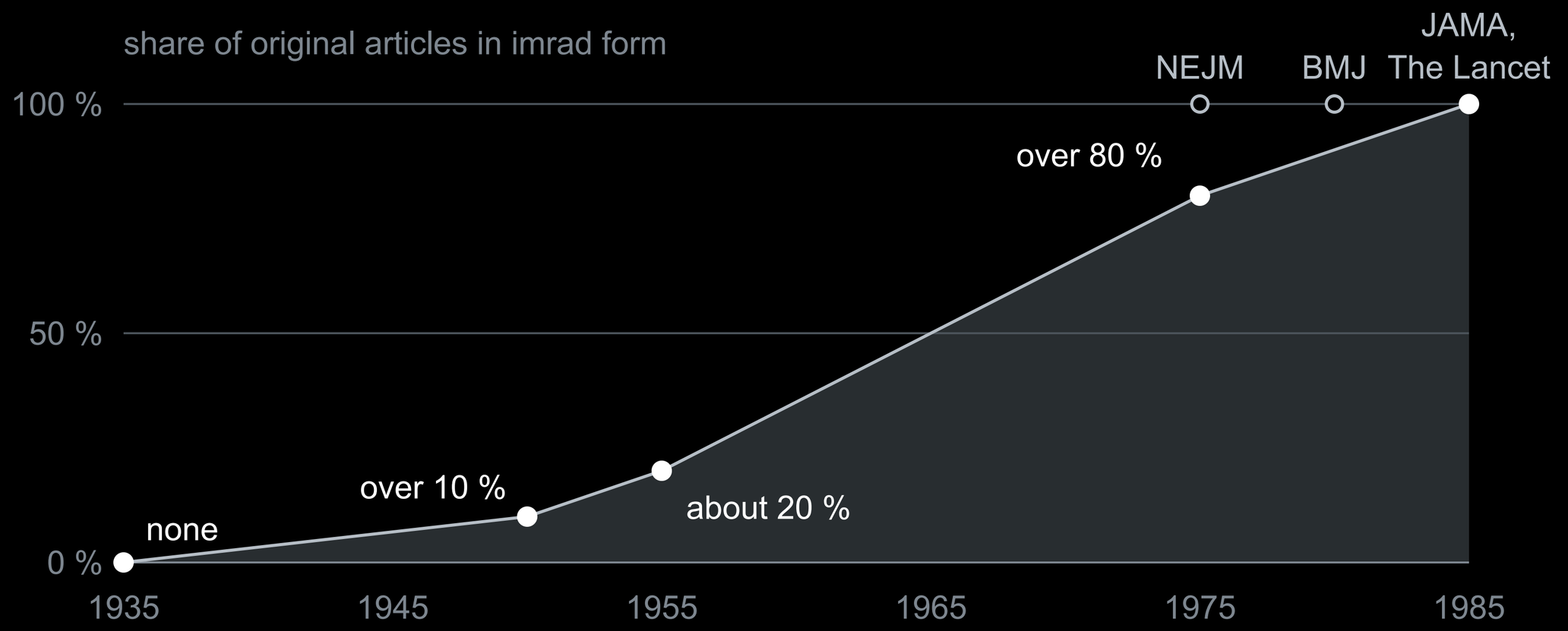


one shape for all papers

imrad



one shape for all papers

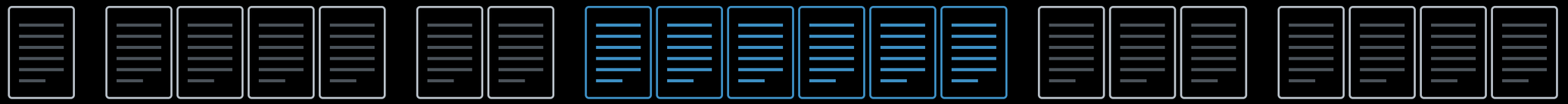


editors made imrad the rule, because readers can then jump straight to the section they need.

Sollaci & Pereira (2004), J Med Libr Assoc 92(3), 364–367: 1,297 original articles in BMJ, JAMA, The Lancet, NEJM

how long is each part?

a rule of thumb for an empirical paper of 20 pages



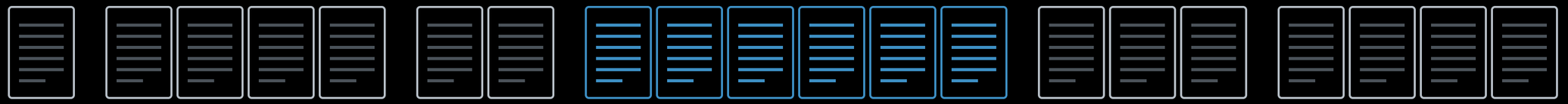
introduction | theory | methods | **results** | discussion | references

the results take the most room, and the theory must not crowd them out.

after Döring (2023), Forschungsmethoden und Evaluation, 6th ed., p. 771

how long is each part?

a rule of thumb for an empirical paper of 20 pages



introduction theory methods results discussion references

and one with twelve pages of theory and two of results, which Döring calls unbalanced

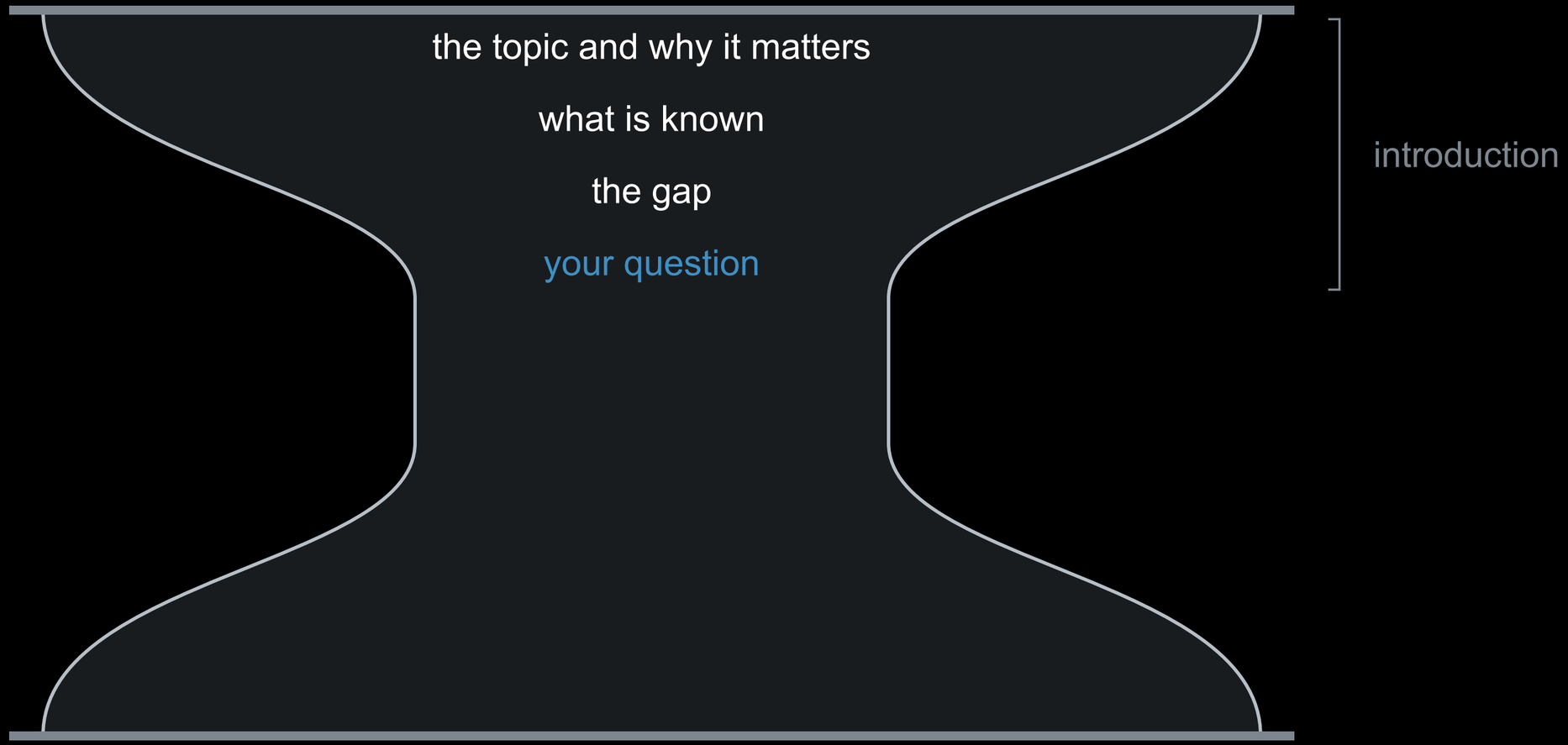


theory results

the results take the most room, and the theory must not crowd them out.

after Döring (2023), Forschungsmethoden und Evaluation, 6th ed., p. 771

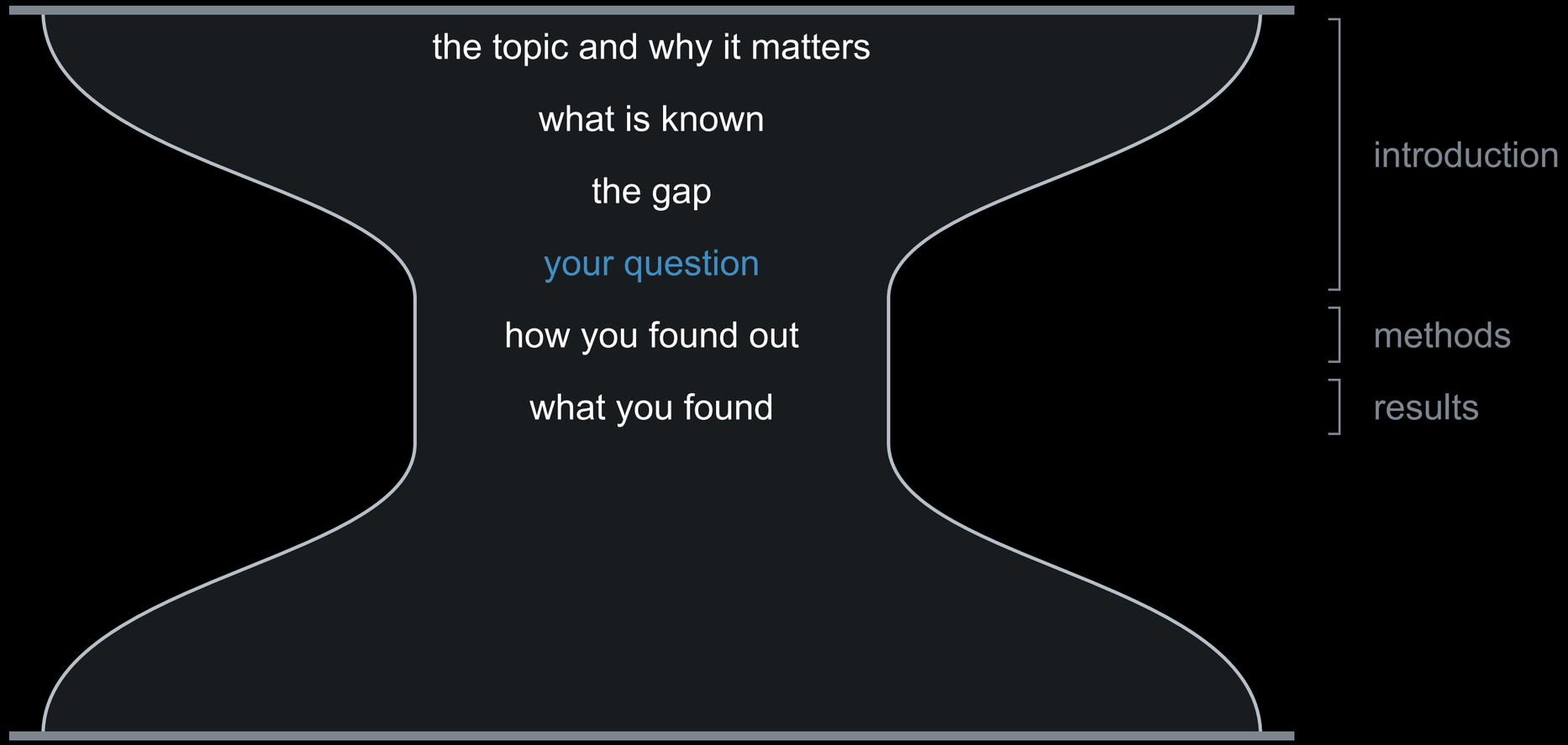
the hourglass



the introduction ends with your question, and the discussion begins with its answer.

after Cals & Kotz (2013), effective writing and publishing scientific papers, J Clin Epidemiol 66

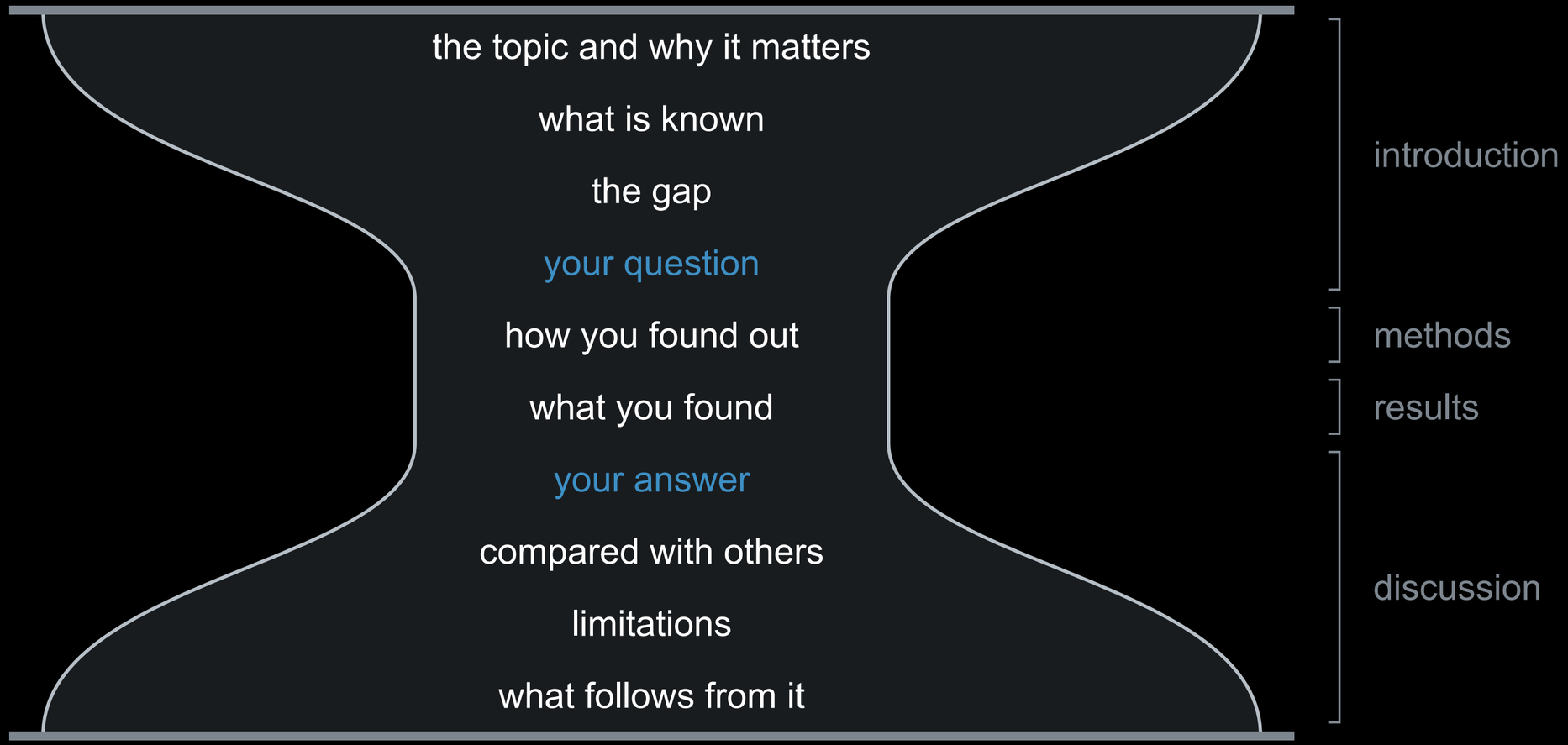
the hourglass



the introduction ends with your question, and the discussion begins with its answer.

after Cals & Kotz (2013), effective writing and publishing scientific papers, J Clin Epidemiol 66

the hourglass



the introduction ends with your question, and the discussion begins with its answer.

after Cals & Kotz (2013), effective writing and publishing scientific papers, J Clin Epidemiol 66

part 2

what goes where

one job for each section

every room of a paper has one job, and what belongs to another room gets carried there.



image: ai-generated (gpt-image-2.5-flare)

1. introduction

milk competes on price in every supermarket.
earlier studies asked why people choose
plant-based milk (Dorina et al., 2025).
but whether price limits differ
with diet is not known.
do price limits for
milk differ by diet?

the introduction narrows from the topic to your question, and the question stands in its last paragraph.

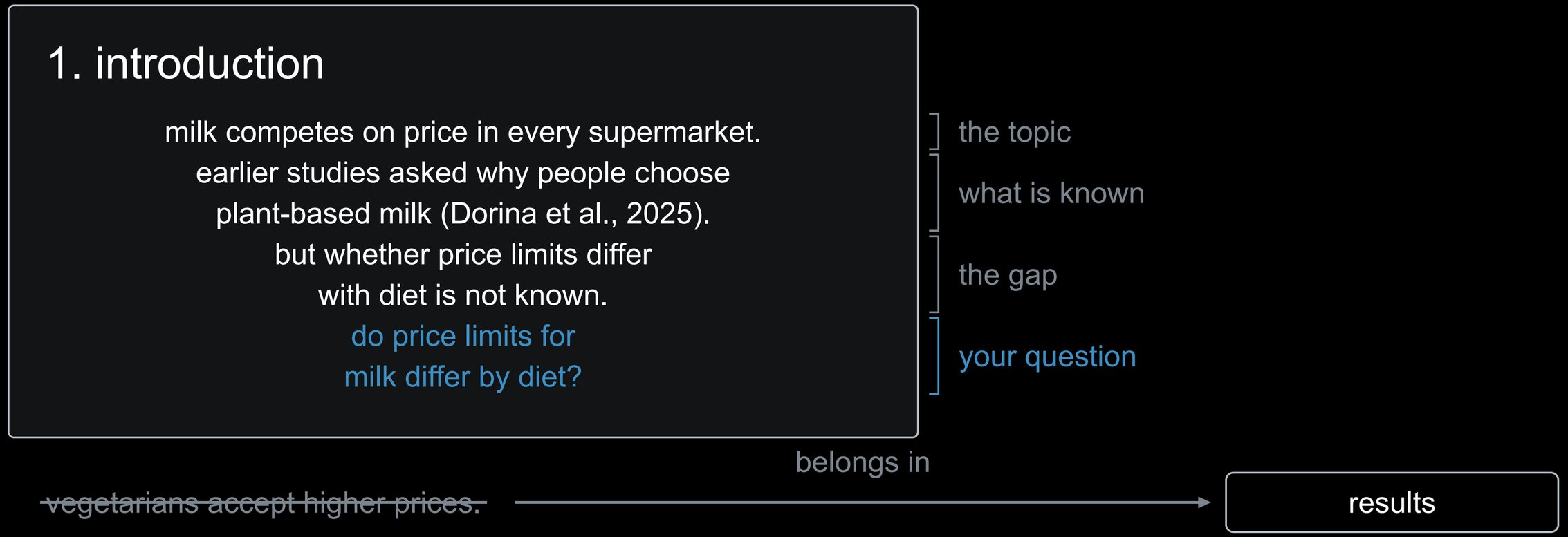
1. introduction

milk competes on price in every supermarket.
earlier studies asked why people choose
plant-based milk (Dorina et al., 2025).
but whether price limits differ
with diet is not known.
do price limits for
milk differ by diet?

- the topic
- what is known
- the gap
- your question

the introduction narrows from the topic to your question, and the question stands in its last paragraph.

example paper written for this input; structure after Cals & Kotz (2013), part III



the introduction narrows from the topic to your question, and the question stands in its last paragraph.

2. methods

we reanalysed the milk and chocolate survey, an existing online survey of 2,809 adults in Germany.
619 respondents saw one of two milks and were asked from which price it would be too expensive; 611 named one.
we left out 23 prices below 0.50 € or above 10 €, which can hardly be meant for a litre, so 588 remained.
we compared the median price limit of the diet groups.

the methods are a recipe that would let someone else repeat your analysis.

example paper written for this input; APA JARS-Quant (2020); Döring (2023), p. 771

2. methods

we reanalysed the milk and chocolate survey, an existing online survey of 2,809 adults in Germany.
619 respondents saw one of two milks and were asked from which price it would be too expensive; 611 named one.
we left out 23 prices below 0.50 € or above 10 €, which can hardly be meant for a litre, so 588 remained.
we compared the median price limit of the diet groups.

- data source
- measure and sample
- cleaning
- analysis

the methods are a recipe that would let someone else repeat your analysis.

example paper written for this input; APA JARS-Quant (2020); Döring (2023), p. 771

2. methods

we reanalysed the milk and chocolate survey, an existing online survey of 2,809 adults in Germany.
619 respondents saw one of two milks and were asked from which price it would be too expensive; 611 named one.
we left out 23 prices below 0.50 € or above 10 €, which can hardly be meant for a litre, so 588 remained.
we compared the median price limit of the diet groups.

- data source
- measure and sample
- cleaning
- analysis

~~the median is the middle value of a sorted list.~~ belongs in → a textbook

the methods are a recipe that would let someone else repeat your analysis.

example paper written for this input; APA JARS-Quant (2020); Döring (2023), p. 771

3. results

vegetarians named a median price limit of 2.45 € (n = 28), omnivores one of 1.99 € (n = 362).
figure 1 shows all five diet groups.



figure 1. median price limit by diet

the results report what the data say, and leave what it means to the discussion.

example paper written for this input, numbers from the compact milk survey; after Döring (2023), p. 772

3. results

vegetarians named a median price limit of 2.45 € (n = 28), omnivores one of 1.99 € (n = 362).
figure 1 shows all five diet groups.



figure 1. median price limit by diet

the finding, with n

pointer to the figure

the full picture

the results report what the data say, and leave what it means to the discussion.

example paper written for this input, numbers from the compact milk survey; after Döring (2023), p. 772

3. results

vegetarians named a median price limit of 2.45 € (n = 28), omnivores one of 1.99 € (n = 362).
figure 1 shows all five diet groups.



figure 1. median price limit by diet

the finding, with n

pointer to the figure

the full picture

belongs in

~~probably because they value animal welfare.~~ → discussion

the results report what the data say, and leave what it means to the discussion.

example paper written for this input, numbers from the compact milk survey; after Döring (2023), p. 772

4. discussion

only vegetarians
named higher price limits.
whether this reflects concern
for animals, the data cannot say.
only 28 vegetarians named a limit, and
each respondent saw only one of two milks.
a larger sample should test whether the difference holds.

the discussion opens with the answer and then widens like an upside-down funnel.

example paper written for this input; after Döring (2023), p. 770, and Cals & Kotz (2013), part VI

4. discussion

only vegetarians
named higher price limits.
whether this reflects concern
for animals, the data cannot say.
only 28 vegetarians named a limit, and
each respondent saw only one of two milks.
a larger sample should test whether the difference holds.

- the answer
- what the data cannot say
- limitations
- what follows

the discussion opens with the answer and then widens like an upside-down funnel.

example paper written for this input; after Döring (2023), p. 770, and Cals & Kotz (2013), part VI

4. discussion

only vegetarians
named higher price limits.
whether this reflects concern
for animals, the data cannot say.
only 28 vegetarians named a limit, and
each respondent saw only one of two milks.
a larger sample should test whether the difference holds.

- the answer
- what the data cannot say
- limitations
- what follows

~~women named higher price limits than men.~~ belongs in → results

the discussion opens with the answer and then widens like an upside-down funnel.

example paper written for this input; after Döring (2023), p. 770, and Cals & Kotz (2013), part VI

part 3

the most common mistake

interpreting in the results

the most common mistake is not a wrong number, but a sentence standing in the wrong room.



image: ai-generated (gpt-image-2.5-flare)

where does this sentence belong? “vegetarians named a higher price limit, probably because they value animal welfare.”

introduction

methods

results

discussion



live.hsos.uber.space/q

where does this sentence belong? “vegetarians named a higher price limit, probably because they value animal welfare.”

- introduction
- methods
- results
- discussion

live.hsos.uber.space/q

where does this sentence belong? “vegetarians named a higher price limit, probably because they value animal welfare.”

- introduction
- methods
- results
- discussion

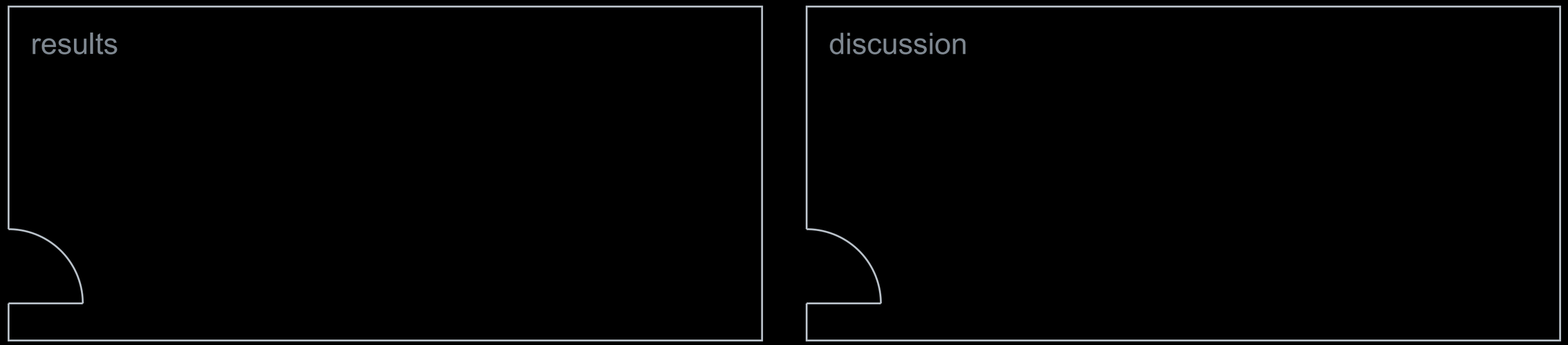
live.hsos.uber.space/q

vegetarians named a higher price limit,
probably because they value animal welfare.

a result says what the data show, and the “because” waits for the discussion.

vegetarians named a higher price limit,

probably because they value animal welfare.



a result says what the data show, and the “because” waits for the discussion.

vegetarians named a higher price limit,

probably because they value animal welfare.

results

vegetarians named a higher price limit than omnivores, a median of 2.45 € against 1.99 € (n = 28 and 362).

discussion

this may reflect that they value animal welfare, a reason the data themselves do not show.

a result says what the data show, and the “because” waits for the discussion.

health, environment or taste?

abstract
1. introduction
    1.1. theory of planned behaviour
    1.2. the current study
2. materials and methods
    2.1. procedure
    2.2. participants
    2.3. measures
    2.4. data analysis
3. results
    3.1. predicting intention
    3.2. predicting plant-based milk consumption
4. discussion
    strengths and limitations
5. conclusions
references

Dorina, Nikpour, Mullan & Uren (2025), Foods 14(11), 1970, CC BY 4.0: section headings of the article

health, environment or taste?

abstract
1. introduction
    1.1. theory of planned behaviour
1   1.2. the current study
2. materials and methods
    2.1. procedure
    2.2. participants
    2.3. measures
    2.4. data analysis
3. results
    3.1. predicting intention
    3.2. predicting plant-based milk consumption
4. discussion
    strengths and limitations
5. conclusions
references

← the question and hypotheses

Dorina, Nikpour, Mullan & Uren (2025), Foods 14(11), 1970, CC BY 4.0: section headings of the article

health, environment or taste?

abstract
1. introduction
    1.1. theory of planned behaviour
1 1.2. the current study
2. materials and methods
    2.1. procedure
    2.2. participants
    2.3. measures
    2.4. data analysis
3. results
    3.1. predicting intention
    3.2. predicting plant-based milk consumption
2 4. discussion
    strengths and limitations
5. conclusions
references

← the question and hypotheses

← the answer, first paragraph

Dorina, Nikpour, Mullan & Uren (2025), Foods 14(11), 1970, CC BY 4.0: section headings of the article

health, environment or taste?

abstract
1. introduction
    1.1. theory of planned behaviour
1 1.2. the current study ← the question and hypotheses
2. materials and methods
    2.1. procedure
    2.2. participants
    2.3. measures
    2.4. data analysis
3. results
    3.1. predicting intention
    3.2. predicting plant-based milk consumption
2 4. discussion ← the answer, first paragraph
3 strengths and limitations ← the limitations
5. conclusions
references

Dorina, Nikpour, Mullan & Uren (2025), Foods 14(11), 1970, CC BY 4.0: section headings of the article

part 4

the abstract

the whole paper in one paragraph

the abstract is the paper's listing: one paragraph that tells a reader whether to open the door.



image: ai-generated (gpt-image-2.5-flare)

introduction

milk competes on price, but whether price limits for milk differ by diet has not been asked.

17 words

the paper in miniature

introduction

milk competes on price, but whether price limits for milk differ by diet has not been asked.

methods

we reanalysed an online survey of 2,809 adults in Germany and compared, across five diets, the price from which 588 of them found a milk too expensive.

44 words

after Döring (2023), p. 776, and the APA publication manual (2020)

the abstract

the paper in miniature

introduction

milk competes on price, but whether price limits for milk differ by diet has not been asked. methods we reanalysed an online survey of 2,809 adults in Germany and compared, across five diets, the price from which 588 of them found a milk too expensive. results vegetarians named a median limit of 2.45 € (n = 28), the other four diets 1.99 € or 2.00 €.

59 words

after Döring (2023), p. 776, and the APA publication manual (2020)

the abstract

the paper in miniature

introduction

milk competes on price, but whether price limits for milk differ by diet has not been asked.

methods

we reanalysed an online survey of 2,809 adults in Germany and compared, across five diets, the price from which 588 of them found a milk too expensive.

results

vegetarians named a median limit of 2.45 € (n = 28), the other four diets 1.99 € or 2.00 €.

discussion

price limits may be higher among vegetarians, which a larger sample should test.

72 words

150 to 250 words are usual, and the abstract is written last

after Döring (2023), p. 776, and the APA publication manual (2020)

the abstract

write your question at the end of the introduction and your answer at the start of the discussion, and let everything else connect the two.



image: ai-generated (gpt-image-2.5-flare)